

Professional ECS/Fargate Deployment Guide for FastAPI IoT Platform

Executive Summary

Recommended when managed infrastructure, autoscaling, and enterprise operations are required.

Recommended Architecture

ALB → ECS Fargate Tasks → Redis → RDS PostgreSQL.

Sizing

2 vCPU / 4 GB RAM and 2 vCPU / 8 GB RAM are common production configurations.

Estimated Monthly Cost

2 vCPU / 4 GB Fargate: ~\$72/month. 2 vCPU / 8 GB Fargate: ~\$85/month. Add RDS: \$30–80/month.

Advantages

No server maintenance, automatic scaling, rolling deployments, multi-AZ support.

Limitations

Higher cost, limited host-level access, additional AWS managed service charges.

Scalability Estimates

500 devices: excellent. 5,000 devices: excellent with autoscaling. 10,000+ devices: supported.

Recommendation

Choose ECS/Fargate if uptime, autoscaling, and operational simplicity are more important than cost.